



Microgrid-Based Energy-Independent Village Planning Through Identification of Rooftop Solar Power Plants Potential in Kedungrong, Yogyakarta



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Abstract: The increasing demand for electrical energy and the commitment to clean energy transition encourage the utilization of renewable energy at the community level through the development of solar photovoltaic (PV). This study aims to identify the potential of solar PV, design the generating system, and analyze the technical and economic feasibility of developing an energy-independent village based on microgrids in Kedungrong Hamlet, Yogyakarta Special Region. The study uses a descriptive quantitative approach with the research object being 53 houses' rooftops identified as having the potential for installing rooftop solar power plants (PLTS). Primary data were obtained through field observations, inventory of roof area and household electricity needs, and technical documentation, while secondary data were obtained from Global Solar Atlas and relevant scientific literature. The analysis was carried out through identification of solar radiation potential, calculation of electricity consumption, design of solar panel capacity and its supporting components, and evaluation of economic feasibility based on initial investment and energy saving benefits. The results of the study indicate that the rooftops of 53 buildings in Kedungrong Hamlet have good potential to be developed as PLTS systems because they are supported by adequate solar radiation intensity throughout the year. Economic analysis shows that PV investments provide benefits in the form of electricity cost savings and have a viable long-term return on investment. The integration of rooftop PV through a microgrid system improves energy supply reliability, electricity distribution efficiency, and strengthens community energy security. This research produces an energy-independent village planning model based on the utilization of 53 rooftops as a source of renewable electricity generation that can be replicated in other rural areas with similar characteristics to support the achievement of the national energy mix and sustainable development.

Keywords: Energy independent village; Rooftop solar photovoltaic; Microgrid; Renewable energy; Economic feasibility analysis

1 Introduction

Indonesia possesses a vast array of renewable energy potential, including solar energy, which is available year-round due to its geographical location on the equator. However, this renewable energy potential has not been optimally and efficiently utilized. The continued high dependence on fossil fuels prevents renewable energy from becoming a primary energy source [1]. The use of renewable energy, particularly solar energy, can provide significant benefits in reducing carbon emissions and mitigating the negative ecological impacts of fossil-based power plants [2]. Integrating renewable energy into power generation systems, including through energy mix schemes, is believed to accelerate the achievement of the national energy transition target for a cleaner, more sustainable, and more environmentally friendly energy source.

Energy is a vital need for human activity across the economic, household, industrial, business, and transportation sectors. Therefore, its availability is a key factor in supporting regional development [3]. Currently, the majority of the world's energy supply still relies on fossil-based energy sources such as oil and coal, which are non-renewable resources and have limited reserves. Energy demand continues to increase in line with population growth, technological developments, and increased economic activity [4].

By utilizing locally available renewable energy resources, a properly designed microgrid can support the development of energy-independent communities by integrating distributed generation, energy storage, electricity loads, and grid interconnection within a defined local electricity network. In Kedungrong, Yogyakarta, rooftop solar photovoltaic (PV) systems represent a promising distributed renewable energy resource because most residential buildings have roofs that receive adequate solar radiation with relatively limited shading from surrounding structures and vegetation. These characteristics provide a suitable basis for developing a low-voltage residential microgrid in which rooftop PV systems can supply household loads, while excess electricity can be managed through battery energy storage and, where technically feasible, shared among connected households. The proposed system considers the low-voltage distribution network as the electrical boundary connecting participating households and the utility grid through a defined point of common coupling (PCC). Within this configuration, an energy management strategy is required to coordinate PV generation, household demand, battery charging and discharging, and electricity exchange with the utility grid. During periods of sufficient solar generation, PV electricity is prioritized for local household consumption, while surplus generation can be stored in the battery or supplied to other connected loads. During periods of insufficient PV generation, stored energy can be dispatched to support local demand, with the utility grid providing supplementary electricity when the available renewable generation and storage capacity are insufficient [5, 6]. This configuration provides a technical framework for assessing the feasibility of developing Kedungrong as an energy-independent village based on a grid-connected rooftop PV microgrid, while the capability for fully islanded operation depends on the implementation of appropriate protection, control, and energy management systems [7].

Rooftop solar power plants (PLTS) demonstrate the technology's significant potential as a technically and economically feasible alternative energy solution. Hasanah [8] designed a 6.4 kWp off-grid PLTS system for a residential home using 32 solar modules, demonstrating that electricity needs can be met independently. Husni et al. [9] also showed that the implementation of a building-scale solar power plant with a capacity of 8,039 Wp provides significant economic benefits and is more optimal when combined with *Perusahaan Listrik Negara* (PLN) and generators in a hybrid system. Meanwhile, Kossi [10] proved that centralized PLTS in remote areas are economically feasible with energy costs that can be reduced through load optimization. Meheng et al. [11] added that the configuration of a hybrid solar power plant with a diesel power plant is the best solution to meet long-term electricity needs in remote areas. This study aims to identify rooftop solar potential, design household-scale systems, estimate installation costs, and develop a strategy for integrating the systems into a microgrid.

2 Literature Review

According to research by Duma et al. [12], utilizing solar energy is an effective way to save electricity consumption. Solar energy can be utilized as an alternative energy source that can reduce dependence on electricity from fossil-fueled power plants [13]. Monitoring of electricity usage and production is carried out in real time via a smartphone application, thus simplifying energy management. The implementation of this research has significant potential to reduce dependence on conventional electricity and support energy sustainability [14]. The development of a microgrid-based system can be a further step in optimizing the collective use of solar energy, thereby supporting the realization of energy-independent villages through more efficient, integrated, and sustainable energy management [15]. This will reduce electricity consumption from State Electricity Company, resulting in a lower monthly electricity bill.

Electricity is a primary need in modern life, as nearly all daily activities depend on its availability. Various household appliances, public facilities, communication systems, and even industrial equipment require electricity as their primary power source to operate optimally [16]. Without an adequate supply of electricity, economic, social, and public services will be disrupted, hampering productivity and the quality of life. Along with technological advancements and the increasing number of electrical appliances, the need for electricity continues to rise. Dependence on electricity extends beyond households to industries, businesses, and public services, all requiring a stable and continuous supply. Reliable availability and efficient management of electricity are crucial for supporting sustainable development and public welfare [17].

PLTS are an environmentally friendly energy solution because their operation produces neither exhaust emissions nor noise pollution. The direct current (DC) electricity generated by solar modules can then be used directly or converted to alternating current (AC) using an inverter, depending on the needs of the electrical equipment used [18]. Solar power systems can be designed in various configurations, such as off-grid systems, making them flexible for application on various scales, from households to large-scale applications. With technological advances and increasing solar module efficiency, PLTS are becoming an increasingly reliable alternative power generator to support sustainable energy supply [19, 20].

Incorrect solar panel installation orientation can lead to a significant decrease in energy production, even if the panel capacity is adequate [21]. In addition to installation direction, the tilt angle of the PV modules also significantly affects the amount of electrical energy generated. The correct tilt angle allows the solar panels to

receive solar radiation perpendicularly, especially during peak hours. Tilt angle adjustments are typically tailored to the geographic location and latitude of the installation site, allowing PV modules to operate more efficiently year-round [22]. Roof configuration analysis and determining the correct orientation angle are crucial aspects of rooftop solar PV system planning to achieve optimal energy output.

A microgrid can be a shopping center, industrial area, campus, or isolated residential area. For its intended purpose, a microgrid is a controllable electrical load system. The load can be constant, increase at night when electricity costs are lower, or even decrease to zero during times of system stress.

3 Research Methods

This research used a descriptive quantitative approach to identify the potential for developing PLTS as the foundation for establishing a microgrid-based energy-independent village in Kedungrong Hamlet, Yogyakarta Special Region. This approach was chosen because it provided a comprehensive overview of solar resource potential, community energy needs, the technical feasibility of the system, and the economic aspects that supported sustainable PLTS implementation.

The research scope covered solar radiation potential, the characteristics of buildings potentially equipped with rooftop PV systems, household electricity consumption, off-grid PV system design, and an analysis of the economic feasibility of system development. Primary data were obtained through field observations, site condition measurements, household electrical equipment inventory, and technical documentation. The rooftop survey was conducted during the field data collection period in Kedungrong, Yogyakarta. Roof shading was assessed through direct visual observation of surrounding trees, buildings, and other potential obstructions. A roof was classified as free from shading when no significant obstruction was observed to block direct sunlight over the main roof area during the survey. Secondary data were obtained from the Global Solar Atlas, the Meteorology, Climatology, and Geophysical Agency (BMKG), PT PLN (Persero), the Indonesian National Standard (*Standar Nasional Indonesia/SNI*), and relevant scientific literature on PLTS technology and microgrid systems.

The solar energy potential analysis was conducted based on the Global Horizontal Irradiance (GHI) value and the geographic conditions of the study area. Next, electrical energy needs were calculated based on the power output of electrical appliances and the duration of use to obtain daily and monthly energy consumption. The results of these calculations are used as the basis for determining the capacity of the solar panels, batteries, solar charge controller (SCC), and inverter according to the electrical load requirements. The system design is carried out using the principle of energy balance so that the energy produced can optimally meet household electricity needs.

Figure 1 illustrates the research workflow for developing a rooftop solar power system aimed at supporting energy self-sufficiency in villages. The study utilizes primary data—specifically rooftop area potential and daily electricity demand—and secondary data regarding solar radiation sourced from the Global Solar Atlas, PLN, and GHI. These data are used to determine the potential installed capacity and daily electricity requirements, as well as the specifications for solar panels, batteries, charge controllers, and inverters. Subsequently, an economic analysis is conducted to assess the initial investment, operational costs, electricity cost efficiency, and the break-even point (BEP), ultimately yielding a rooftop solar power system model suitable for an energy-independent village.

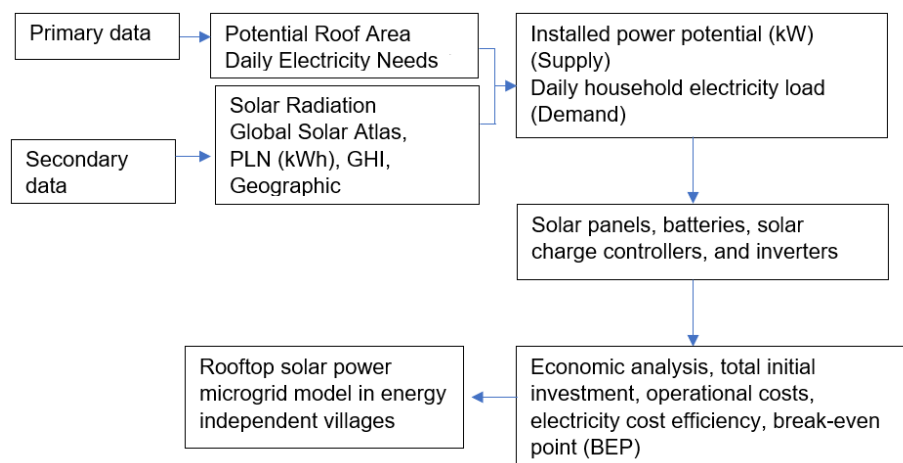


Figure 1. Research flow

Note: PLN = Perusahaan Listrik Negara; GHI = Global Horizontal Irradiance.

An economic analysis was conducted by calculating the total initial investment, operational costs, potential electricity savings, and payback period. All analysis results were then integrated to evaluate the feasibility of

implementing PLTS in a microgrid scheme as a model for energy-independent villages. This approach yields technical and economic recommendations that can serve as a reference for developing community-based renewable energy systems in rural areas and supporting the achievement of a sustainable energy transition.

4 Research Results

GHI data is a crucial parameter indicating the amount of solar radiation received on a horizontal surface, expressed in kWh/m²/day. This information is widely used as a reference in the planning and analysis of PLTS potential because it accurately reflects the availability of solar energy in a given area as explained by Darmawan et al. [23]. National Aeronautics and Space Administration (NASA) prediction data is a reliable reference for determining solar power generation potential, particularly during site identification and capacity planning stages.

The relatively stable solar radiation potential throughout the year allows solar power plant systems to generate electricity optimally and sustainably. Therefore, NASA served as the basis for this study to identify solar energy potential in Kedungrong Village, Yogyakarta. This radiation value was then used as the primary parameter in calculating the electrical energy potential, planning the capacity of PLTS, analyzing technical feasibility, and as the basis for developing a microgrid system towards achieving an Energy Independent Village.

Table 1 shows that the average solar irradiance GHI in the Special Region of Yogyakarta Province peaked in September, with a value of 5.540 kWh/m²/day and a clearness index of 0.545. Meanwhile, January recorded the lowest average solar irradiance and clearness index, with values of 4.280 kWh/m²/day and 0.396, respectively. Overall, the annual average solar irradiance reached 4.80 kWh/m²/day. It is also noteworthy that August had the highest clearness index, at 0.563, while September had the highest average solar irradiance.

Table 1. Solar radiation in Yogyakarta

Month	Clearness Index (Dimensionless)	Mean Daily Solar Irradiation (kWh/m ² /day)
January	0.396	4.280
February	0.413	4.470
March	0.437	4.590
April	0.484	4.720
May	0.533	4.730
June	0.542	4.550
July	0.559	4.800
August	0.563	5.250
September	0.545	5.540
October	0.506	5.390
November	0.438	4.710
December	0.426	4.570

GHI is a parameter that indicates the amount of solar irradiance received by a horizontal surface at the Earth's surface. Solar irradiance represents the power of solar radiation received per unit area and is generally expressed in watts per square meter (W/m²). When accumulated over a day, solar radiation is expressed as daily solar irradiation, with units of kilowatt-hours per square meter per day (kWh/m²/day) [24].

As shown in Table 2, the results of the identification of the characteristics of the research sites resulted in 53 rooftop units being analyzed in the planning of the PLTS. All study sites had relatively uniform solar radiation values, namely 4.80 kWh/m²/day, indicating that the study area has good solar energy potential for use as a power generation source. This uniformity of radiation intensity provides an advantage in the planning process because the basic capacity of the PLTS system can be designed with the assumption of nearly the same radiation potential at each location. Furthermore, the results of the field survey showed that 100% of the locations experienced no shadow interference from surrounding buildings or vegetation. This condition greatly supports increasing the energy conversion efficiency of PV modules because the solar panels can receive optimal sunlight exposure without experiencing a decrease in energy production due to the shadow effect. The measurements also showed that roof areas have quite large variations, with a minimum area of 26.00 m² on Roof Number 53 and a maximum area of 355.61 m² on Roof Number 45. The difference in area provides flexibility in determining the installation capacity of solar modules according to the characteristics of each building. The electrical energy generated by the PV system is stored in a battery and then converted from DC to AC using a 1,000 W inverter. The system is also equipped with a Miniature Circuit Breaker (MCB) to provide protection against electrical faults. In addition, a Maximum Power Point Tracking (MPPT) charge controller is used to optimize the power generated by the solar modules before it is stored in the battery. The inverter plays a crucial role because most household electrical appliances operate using AC at a standard voltage of around 220 V. With these components, the electrical energy generated by the solar panels can

be directly utilized to meet household electricity needs, such as lighting, charging electronic devices, and operating low- to medium-power household appliances [10].

Table 2. Site characteristics relevant to rooftop solar development

No.	Parameters	Value
1	Number of research locations	53 Rooftop units
2	Average solar radiation	4.80 kWh/m ² /day
3	Shading conditions	All locations without shade (100%)
4	Minimum roof area	26.00 m ² (Roof No. 53)
5	Maximum roof area	355.61 m ² (Roof No. 45)
6	Potential for solar power plant installation	All homes are suitable for rooftop solar installation

This system is also equipped with a MCB, which functions as a protective device to protect the electrical installation from the risk of overcurrent or short circuits. The MCB is installed on the electrical distribution line between the inverter and the load so that if a system failure occurs, the power supply can be automatically cut off. The use of this safety system is crucial for maintaining the safety of the solar power plant installation while also improving the system’s long-term reliability.

Figure 2 illustrates the basic configuration of an off-grid rooftop solar PV system designed to independently meet a portion of a household’s electricity needs. This configuration serves as the baseline system for the design and implementation of household-scale solar PV systems in Kedungrong, Yogyakarta. The system comprises a 300 Wp PV module, a 30 A/12 V SCC, a 12 V/100 Ah battery, and a 1,000 W/12 V pure sine wave (PSW) inverter. The PV module converts solar radiation into DC electrical energy. This energy is then routed through the SCC, which regulates the battery charging process and maintains the stability of the energy storage system. The 12 V/100 Ah battery stores the energy generated by the solar PV system for use when PV output is insufficient to meet the load demand, while the 1,000 W/12 V inverter converts DC electrical energy from the battery into AC to power household appliances. Figure 2 thus represents the basic household solar PV system used as a technical reference; meanwhile, the larger system capacities used in the potential and economic analyses are determined separately based on the specific energy requirements and calculation results for each household. This distinction between the basic configuration shown in Figure 2 and the calculated configurations ensures that each design stage has a clear function and calculation basis, while also ensuring reproducibility.

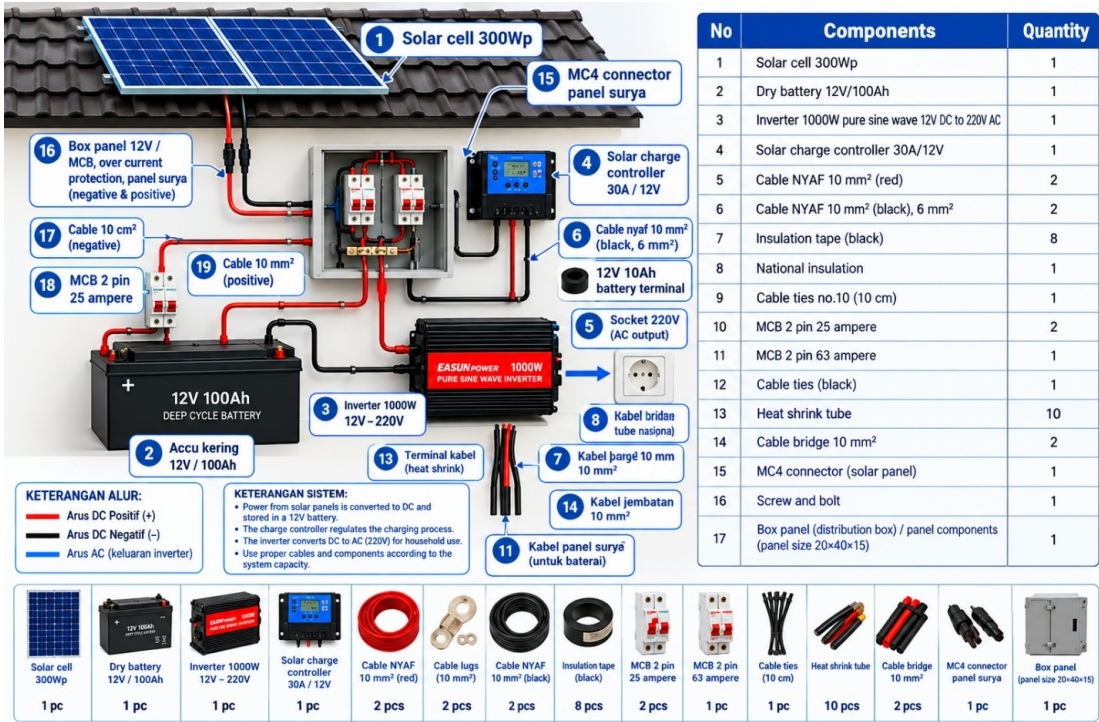


Figure 2. Rooftop solar power plants (PLTS) planning design

Based on the mapping results (Figure 3), a design was obtained for the location planning for the installation of

PLTS in 53 houses in Kedungrong Village, Kulon Progo Regency, Yogyakarta Special Region. Each house has been given an identification number to facilitate the inventory process, roof area measurement, coordinate determination, and analysis of the potential for solar panel installation. The map shows that all research objects are located in a single residential area connected by a village road network, thus facilitating the field survey process and planning the installation of the rooftop solar power system. The location identification results show that most buildings are located in open areas and are not obstructed by tall buildings or vegetation that could potentially cast shadows on the roof.

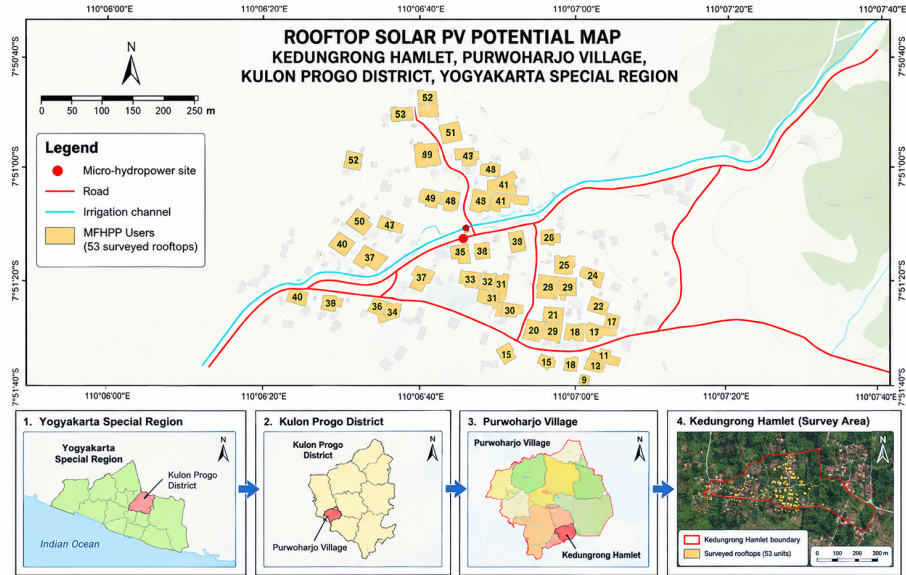


Figure 3. Site plan for the 53 surveyed rooftops in Kedungrong Hamlet, Purwoharjo Village, Kulon Progo District, Yogyakarta Special Region.
 Note: Base map of Yogyakarta Special Region, Kulon Progo District, and Purwoharjo Village obtained from Ina-Geoportal (<https://tanahair.indonesia.go.id/>); satellite imagery from Google Earth (© 2024 Google). The survey overlay (rooftop locations and MFHPP users) was conducted by the authors.

Figure 3. Site plan for the 53 surveyed rooftops

Based on the analysis of battery capacity and load patterns, the PLTS system is capable of supplying lighting loads for 12 and 5 hours consistently. The use of lithium iron phosphate (LiFePO_4) batteries offers advantages in terms of efficiency, lifespan, and safety compared to conventional batteries. According to research by Purnomo and Nugroho [24], an economic analysis was conducted to assess the feasibility of implementing a 1,200 Wp solar power system in households, in terms of investment costs and electricity savings. The basic assumptions adopted in this economic analysis are summarized in Table 3.

Based on the calculation of the installation costs for a residential rooftop solar system, the total investment required to build the system is IDR 3,859,500.00. This cost includes the procurement of various key components required for the solar system, such as a 300 Wp solar panel, a 12 V/100 Ah dry-cell battery, a 1,000 watt/12 V inverter, and a 30 A/12 V SCC to regulate the charging process from the solar panel to the battery. These key components are the core of the solar system, playing a role in converting solar energy into usable electrical energy to meet household electricity needs.

In addition to these main components, the installation cost also includes several supporting components, such as various sizes of electrical cables, solar panel connectors (MC4), DC MCBs for safety, a panel box for component placement, and various installation equipment such as insulation, cable clamps, nuts, bolts, and cable ties. These supporting components serve to ensure the solar system is properly installed, safe, and capable of optimal long-term operation. With a relatively affordable total investment cost, this household-scale rooftop solar power system shows quite good potential to be implemented as an alternative source of environmentally friendly and sustainable electrical energy.

The economic analysis shows that the 1,200 Wp rooftop solar PV system designed for the second household has an estimated payback period of approximately 13.17 years. This value is calculated based on the total initial investment of IDR 25,000,000 and estimated annual electricity savings of approximately IDR 1,898,336. The resulting payback period remains below the expected 20–25-year operational lifetime of solar PV modules, indicating that the proposed system has the potential to be economically feasible for residential application. The total estimated initial investment for this configuration is IDR 25,000,000, including the PV modules, MPPT SCC, LiFePO_4 battery, inverter, cables, and protection equipment. Therefore, the proposed 1,200 Wp system has the potential to reduce household electricity expenditure and provide a reasonable economic return over its operational lifetime.

Based on the technical and economic analysis, the 1,200 Wp rooftop solar PV system presented in Table 3 is considered suitable for the electricity requirements of the second household. The system consists of 12 PV modules with a capacity of 100 Wp each, resulting in a total installed capacity of 1,200 Wp, combined with a 60 A MPPT SCC, a 12 V/400 Ah LiFePO₄ battery, and a 300 W PSW inverter. Under the estimated solar radiation conditions, the system is capable of producing approximately 3.6 kWh of electricity per day, while the household lighting load requires approximately 2.15 kWh per day. The 12 V/400 Ah LiFePO₄ battery provides energy storage to support household loads during periods when solar generation is insufficient. It should be emphasized that this configuration represents a separate household case from the 300 Wp system presented in Figure 2. The difference in system capacity reflects the different electricity requirements and design characteristics of the respective households rather than alternative configurations or sequential design stages of the same system.

Table 3. Cost comparison of the existing 300 Wp and previous 1,200 Wp rooftop solar PV systems

No.	Component	Specification	Quantity/Unit	Unit Price (IDR)	Subtotal / Estimated Cost (IDR)
System Existing 300 Wp					
1	Solar panel	300 Wp	1 unit	1,350,000.00	1,350,000.00
2	Dry battery	12 V/100 Ah	1 unit	1,100,000.00	1,100,000.00
3	Taffware inverter	1,000 W/12 V PSW	1 unit	575,000.00	575,000.00
4	Solar charge controller (SCC)	30 A/12 V	1 unit	75,000.00	75,000.00
5	Red NYAF cable	10 mm ²	2 m	37,500.00	75,000.00
6	SKUN SC25-8	Grade A [JM25-8]	2 units	3,000.00	6,000.00
7	Federal black NYAF cable	6 mm ²	8 m	17,000.00	136,000.00
8	National black insulation tape	—	1 unit	8,000.00	8,000.00
9	Yans concrete cable clamps	No. 10	1 pack	10,000.00	10,000.00
10	DC MCB, 2-pole	25 A	2 units	72,000.00	144,000.00
11	DC MCB, 2-pole	63 A	1 unit	72,000.00	72,000.00
12	Cable ties	—	1 pack	10,000.00	10,000.00
13	Small bolt and nut	—	10 sets	300.00	3,000.00
14	Jembo cable	10 mm ²	2 m	33,000.00	66,000.00
15	MC4 solar panel connector	—	1 set	8,000.00	8,000.00
16	Bolt	—	1 unit	1,500.00	1,500.00
17	TAV panel box	30 × 40 × 15 cm	1 unit	220,000.00	220,000.00
Total Investment—Existing 300 Wp					3,859,500.00
System Previous 1,200 Wp					
18	Solar panels	12 × 100 Wp	12 units	—	12,000,000.00
19	SCC	MPPT 60 A	1 unit	—	2,500,000.00
20	Battery	LiFePO ₄ 12 V/400 Ah	1 unit	—	7,500,000.00
21	Inverter	Pure sine wave (PSW), 300 W	1 unit	—	1,500,000.00
22	Cables and protection	MCB, DC fuse, panel box	1 set	—	1,500,000.00
Total Investment—Previous 1,200 Wp					25,000,000.00

Note: DC = direct current; MCB = miniature circuit breaker; — indicates “not applicable” in the Specification column and “data not available” in the Unit Price column.

Table 4 shows that the research was conducted on 53 rooftop units that were the objects of analysis in the planning of a PLTS. The analysis of energy needs and electricity bills was conducted based on electricity usage data from January to December, thus providing a snapshot of household energy consumption patterns over the course of a year. The calculations show that House No. 28 has the highest energy demand, at 35.93 kWh per day, equivalent to 1,077.90 kWh per month. This high energy demand is influenced by the number and type of electrical appliances

used, although some of its energy needs are supplied by the Micro-Hydro Power Plant (PLTMH) system, resulting in relatively lower electricity consumption from the PLN grid compared to its actual energy needs.

Table 4. Household energy needs and electricity bills

No.	Parameter	Value
1	Number of buildings	53 buildings
2	Houses using PLN electricity	48 houses
3	Bill observation period	January–December (12 months)
4	Maximum daily energy demand	35.93 kWh/day (House No. 28)
5	Maximum monthly energy demand	1,077.90 kWh/month (House No. 28)
6	Highest average PLN electricity bill	Approximately IDR 180,000/month (household ID to be verified against Appendix 4)

Note: PLN = Perusahaan Listrik Negara.

Based on Table 5, the analysis of rooftop solar PV potential indicates that the houses in Kedungrong have considerable potential for rooftop solar energy development. The initial theoretical potential was estimated using the equation below, where (A) represents the total roof area, (G) is the average solar radiation of 4.8 kWh/m²/day, and eta represents the PV module efficiency of 15%. The output EPLTS is expressed in kWh/day. However, the total roof area cannot be assumed to be fully available for PV installation because parts of the roof may be occupied by ridges, edges, access areas, unsuitable surfaces, spacing between modules, and structural limitations. Therefore, the theoretical values obtained from the total roof area, ranging from 18.72 kWh/day for the smallest roof area of 26.00 m² to 256.04 kWh/day for the largest roof area of 355.61 m², should be interpreted as an upper-bound estimate rather than the actual electricity production of the proposed systems.

Table 5. Analysis of electrical energy potential and energy surplus of rooftop solar power plants (PLTS)

No.	Roof Area (m ²)	Solar Power Potential (kWh/day)	Requirements (kWh/day)	Energy Surplus (kWh/day)	Information	Building Type
1	27.12	19.53	0.00	19.53	Surplus	Power House
2	243.49	175.31	24.59	150.72	Fulfilled	Home
3	273.14	196.66	17.53	179.13	Fulfilled	Home
4	284.61	204.92	26.21	178.71	Fulfilled	Home
5	112.68	81.13	17.64	63.49	Fulfilled	Home
6	36.77	26.47	0.00	26.47	Surplus	Home
7	159.26	114.67	23.98	90.69	Fulfilled	Home
8	195.77	140.95	10.49	130.46	Fulfilled	Home
9	144.68	104.17	17.53	86.64	Fulfilled	Home
10	104.27	75.07	16.93	58.14	Fulfilled	Home
11	94.55	68.08	16.49	51.59	Fulfilled	Home
12	71.80	51.70	18.42	33.28	Fulfilled	Home
13	350.82	252.59	10.39	242.20	Fulfilled	Home
14	217.99	156.95	16.93	140.02	Fulfilled	Home
15	181.10	130.39	11.03	119.36	Fulfilled	Home
16	149.43	107.59	11.54	96.05	Fulfilled	Home
17	345.33	248.64	26.41	222.23	Fulfilled	Home
18	232.06	167.08	18.05	149.03	Fulfilled	Home
19	237.87	171.27	23.17	148.10	Fulfilled	Home
20	89.55	64.48	15.87	48.61	Fulfilled	Home
21	87.98	63.35	24.22	39.13	Fulfilled	Home
22	45.84	33.00	17.54	15.46	Fulfilled	Home
23	39.31	28.30	16.12	12.18	Fulfilled	Home
24	90.47	65.14	18.94	46.20	Fulfilled	Home
25	72.30	52.06	16.73	35.33	Fulfilled	Home

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26	93.46	67.29	17.89	49.40	Fulfilled	Home
27	217.33	156.48	16.67	139.81	Fulfilled	Home
28	197.62	142.29	35.93	106.36	Fulfilled	Home
29	98.21	70.71	18.74	51.97	Fulfilled	Home
30	139.91	100.74	16.91	83.83	Fulfilled	Home
31	151.56	109.12	24.64	84.48	Fulfilled	Home
32	147.55	106.24	17.68	88.56	Fulfilled	Home
33	188.31	135.58	24.96	110.62	Fulfilled	Home
34	152.64	109.90	17.61	92.29	Fulfilled	Home
35	277.47	199.78	18.60	181.18	Fulfilled	Home
36	186.78	134.48	24.15	110.33	Fulfilled	Home
37	163.87	117.99	18.00	99.99	Fulfilled	Home
38	197.17	141.96	18.90	123.06	Fulfilled	Home
39	188.63	135.81	17.58	118.23	Surplus	cowshed
40	168.66	121.44	17.94	103.50	Fulfilled	Home
41	51.49	37.07	24.52	12.55	Fulfilled	Home
42	146.45	105.44	20.46	84.98	Fulfilled	Home
43	175.74	126.53	18.39	108.14	Fulfilled	Home
44	226.80	163.30	20.68	142.62	Fulfilled	Home
45	355.61	256.04	19.83	236.21	Fulfilled	Home
46	252.99	182.15	15.75	166.40	Fulfilled	Home
47	118.68	85.45	17.97	67.48	Fulfilled	Home
48	138.92	100.02	17.46	82.56	Fulfilled	Home
49	44.80	32.26	0.00	32.26	Surplus	cowshed
50	56.68	40.81	0.00	40.81	Surplus	cowshed
51	131.58	94.74	18.58	76.16	Fulfilled	Home
52	106.14	76.42	16.39	60.03	Fulfilled	Home
53	26.00	18.72	0.00	18.72	Surplus	cowshed

$$E_{PLTS} = A \times G \times \eta$$

$$A = \text{Roof area (m}^2\text{)}$$

$$G = \text{Solar radiation (4.8 kWh/m}^2\text{/day)}$$

$$H = \text{Panel efficiency (15\% = 0.15)}$$

$$(t) = 5 \text{ hours/day}$$

$$P = \frac{E_{PLTS}}{t}$$

$$P = \text{kW}$$

$$E_{PLTS} = A \times G \times \eta \quad (1)$$

From an environmental perspective, the electricity generated by the proposed rooftop solar PV system can displace a portion of the electricity supplied by the conventional grid. The amount of grid electricity displaced is based on the solar PV system's annual energy production, while the resulting CO₂ emission reduction is calculated using the formula CO₂ = E/EF . The grid emission factor used in this study is 0.697 kg CO₂/kWh for the Java-Bali electricity system, based on data reported by Indonesia's Ministry of Energy and Mineral Resources (ESDM).

To obtain a more realistic estimation, the technically usable roof area should be determined by applying a roof utilization factor that accounts for installation constraints. In addition, the actual electricity delivered by the PV system is affected by system losses, including temperature effects, inverter conversion losses, cable losses, soiling, mismatch losses, and battery charging and discharging losses. Accordingly, the practical energy output can be expressed as the equation below, where f represents the proportion of roof area that can realistically be occupied by PV modules and PR represents the overall system performance ratio. This approach provides a more conservative estimation of the electricity that can actually be generated by the proposed rooftop PV systems.

$$E_{\text{real}} = A \times f_{\text{usable}} \times G \times \eta \times PR \quad (2)$$

The revised assessment indicates that the available rooftop area and solar radiation provide a substantial opportunity for rooftop PV development in Kedungrong. Nevertheless, the amount of electricity that can be generated and the resulting energy surplus vary among households depending on the usable roof area, installed PV capacity, household electricity demand, system losses, and energy storage characteristics. The potential energy values presented in Table 5 should not be interpreted as evidence that every household will necessarily generate a

substantial energy surplus. Instead, they represent the theoretical and technically constrained potential that can be used as a basis for determining the appropriate PV capacity for each household. This more realistic assessment also provides a stronger technical basis for developing a residential rooftop PV microgrid, in which electricity generation, household demand, energy storage, and electricity exchange among participating households can be evaluated according to the actual capacity of each system.

The energy requirement calculations presented in Table 5 were derived from an inventory of electrical appliances and estimated usage durations based on observations and interviews with the residents. These figures represent the estimated total household energy consumption, whereas the planned solar PV capacity is intended to meet only a portion of the demand—specifically, priority loads. Consequently, the 1.2 kWp system at House No. 28—which generates approximately 3.42–3.60 kWh/day—is not designed to cover the total consumption of 35.93 kWh/day, but rather to serve as an energy source for priority loads within the microgrid system. Overall, the residential rooftop solar power system design demonstrates a simple yet effective configuration for generating and managing electricity from a renewable energy source. The integration of solar panels, a charge controller, a battery, an inverter, and safety systems enables optimal operation in providing alternative electrical energy for household needs. With a design that is relatively simple and easy to implement, this system holds significant potential to support independent solar energy utilization while reducing reliance on electricity from the conventional grid.

As shown in Figure 4, the proposed residential microgrid in Kedungrong is designed as a hybrid renewable energy system integrating 53 rooftop PV units, battery energy storage systems, household priority loads, a micro-hydropower plant (PLTMH), and a low-voltage distribution network. Each household is equipped with a rooftop PV system and battery storage to supply its priority electrical loads. All household systems are interconnected through a low-voltage distribution network and coordinated at a PCC, allowing electricity generated by individual PV systems to be shared among participating households. The PLTMH is integrated into the same network as a complementary generation source, providing electricity when solar irradiation is low or PV generation is insufficient. The hybrid configuration therefore combines the distributed nature of PV generation with the more continuous generation capability of the PLTMH, while battery storage provides flexibility in balancing generation and household demand.

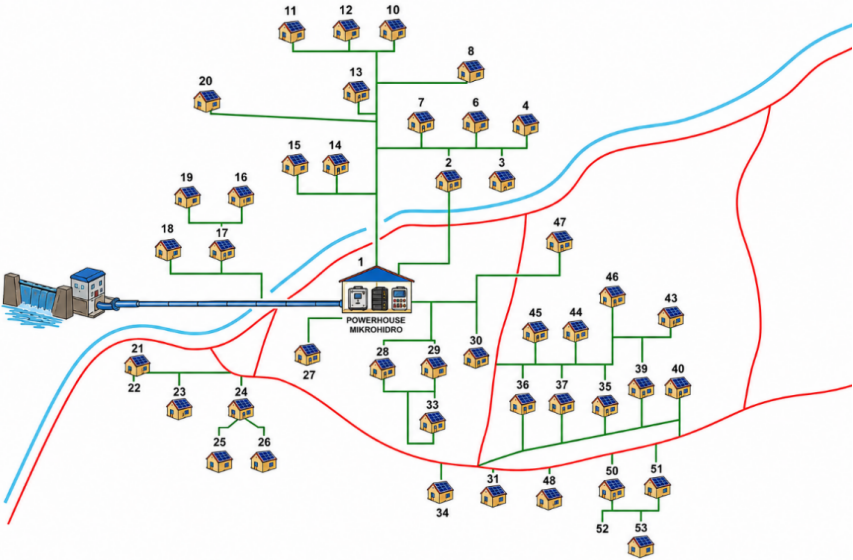


Figure 4. Micro-hydro microgrid planning design connecting the 53 surveyed rooftops in Kedungrong

The energy management principle is based on prioritizing local household consumption, followed by battery charging and electricity sharing within the microgrid. When rooftop PV generation exceeds the instantaneous demand of a household, the surplus electricity is first directed to battery storage. Once the battery reaches its allowable charging level, the remaining surplus can be supplied through the low-voltage network to other households experiencing higher demand. Conversely, when PV generation is insufficient, the household load can be supported by the stored battery energy and PLTMH generation through the common network. The PLN grid can function as a supplementary source when the available renewable generation and stored energy are insufficient, depending on the final interconnection arrangement. This hybrid operating concept enables electricity exchange among households, improves the utilization of locally generated renewable energy, reduces dependence on conventional grid electricity, and provides a more flexible and reliable energy supply for the development of a microgrid-based energy-independent village.

5 Discussion

The microgrid development strategy developed in this study begins with the installation of rooftop solar power systems at each house, based on potential identification results. The individual rooftop PV systems can subsequently be interconnected through the existing low-voltage distribution network to form an integrated community microgrid [25]. Within this configuration, surplus PV electricity may be shared among participating households, subject to network constraints and an appropriate energy-management mechanism [26]. In the next stage, the electrical energy generated by each house can be used primarily for its own needs, while excess energy can be channeled to the microgrid network for use by other homes in need. This strategy allows for more efficient solar energy utilization because energy production can be shared within a single area [5, 27].

The results of the strategy development also indicate that the presence of an energy storage system in the form of a battery in each home plays a crucial role in maintaining a continuous electricity supply. Energy generated during the day is stored in the battery and used when solar radiation intensity decreases or at night [28]. If the microgrid system is further developed with integrated energy management, the reliability of electricity supply will increase; energy distribution will become more equitable, and solar energy utilization will be sustainable. This allows the system to function not only as a household power generator but also as a mutually supportive community energy network. Overall, the research results indicate that Kedungrong Village has excellent potential for developing a rooftop solar-based microgrid [29]. High solar radiation potential, available roof space, the close distribution of houses, and feasible technical and economic planning are supporting factors in developing this development strategy.

Measurements show that each house receives relatively even solar radiation exposure and has an open roof without significant shadow interference. These conditions are key assets in developing a microgrid development strategy, as they enable the formation of a local electricity distribution network with relatively short distances between houses. The technical design results for PLTS indicate that each home can be equipped with a solar power generation system based on the available roof area [30, 31]. The system design consists of solar panels, a SCC, batteries, an inverter, and protection devices capable of generating and storing electrical energy to meet household needs.

The economic analysis indicates that PLTS are feasible as a long-term energy investment. The use of solar energy can reduce electricity consumption from the conventional grid, thereby providing savings on electricity costs for the community. This technical and economic assessment supports the potential development of microgrids with long-term operational and investment benefits, while retaining the study's limitations.

6 Conclusion

This study confirms that the 53 houses in Kedungrong Hamlet, Yogyakarta Special Region, have significant potential for rooftop PV development, supported by an average solar irradiation of 4.80 kWh/m²/day and roof areas ranging from 26.00 m² to 355.61 m². However, the theoretical potential should not be interpreted as actual electricity production because not all roof surfaces are suitable for PV installation. Roof edges, access areas, module spacing, orientation, and structural conditions can reduce the effective installation area. Therefore, an effective roof-area factor provides a more realistic basis for determining the appropriate PV capacity for each household. The estimated total household electricity demand reaches 912.98 kWh/day, demonstrating substantial variation in household load profiles. Consequently, PV system sizing should consider individual electricity demand and priority loads rather than roof area alone. In the economic scenario, a 1.2 kWp rooftop PV system is estimated to generate approximately 1,314 kWh/year, requiring an initial investment of IDR 25,000,000, with estimated annual electricity savings of IDR 1,898,336 and a simple payback period of approximately 13.17 years. The annual electricity savings were calculated using an electricity tariff of IDR 1,444.70/kWh, corresponding to the tariff applied in the economic scenario. This tariff was used consistently to estimate the annual electricity savings and was not based on the IDR 1,467.28/kWh tariff reported in the literature review. These results indicate that rooftop PV can provide long-term economic benefits while supporting local energy security and reducing dependence on conventional grid electricity. The proposed energy-independent village concept is designed as a hybrid residential microgrid integrating rooftop PV systems, battery storage, household priority loads, PLTMH, a low-voltage distribution network, and a common connection point. Surplus PV electricity can be prioritized for battery charging and subsequently shared with households experiencing higher demand, while PLTMH can support the system when solar generation is insufficient. Nevertheless, this microgrid concept has not yet been validated through detailed power-flow analysis, protection coordination, control-system testing, structural assessment, or field-scale operation. The study is limited to one village and uses average solar irradiation and assumed component performance. Household loads may vary with household characteristics, appliance ownership, daily activities, and seasonal patterns. Component aging, PV degradation, battery replacement, and maintenance costs may also affect long-term performance and economic feasibility. Seasonal shading variations were not specifically assessed. Therefore, implementation in other rural areas requires adequate solar resources, structurally suitable rooftops, reliable renewable energy sources, appropriate low-voltage infrastructure, sufficient energy storage, and local capacity for operation and maintenance. Further technical, environmental, and economic validation is required before full-scale implementation.

Author Contributions

Conceptualization, A.A. and A.K.; methodology, A.A.; software, A.A.; validation, A.A., M.A., A.K., and D.T.R.; formal analysis, A.A. and A.K.; resources, A.A.; data curation, A.A.; writing—original draft preparation, A.A. and M.A.; writing—review and editing, M.A. and D.T.R.; visualization, A.A., M.A., A.K., and D.T.R.; supervision, A.A., M.A., A.K., and D.T.R.; project administration, A.A.; digital marketing, A.A.; community empowerment, A.A. All authors have read and agreed to the published version of the manuscript.

Data Availability

The data used to support the research findings are available from the corresponding author upon request.

Conflicts of Interest

The authors declare no conflicts of interest.

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